Donghyeon Kim

Molecular Machine Learning · Computational Biology and Chemistry

Integrated M.S.–Ph.D. Student, Bio and Brain Engineering, KAIST

kdysmurf@kaist.ac.kr
+82 10-6653-8377 · Daejeon, South Korea
dhyeonk-0008.github.io
github.com/dhyeonk-0008 · LinkedIn

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Feb 2025 – Present
Integrated M.S.–Ph.D. | GPA 3.98 / 4.3 | Expected graduation: Feb 2030 | Advisor: Prof. Dongsup Kim — Daejeon, South Korea

Seoul National University (SNU) — Mar 2020 – Feb 2025
B.S. in Chemistry Education | GPA 3.31 / 4.3 — Seoul, South Korea

RESEARCH PROJECTS

DeMiR: Density-Matrix-Informed Representations for Molecular Property Prediction — May 2025 – Present
Quantum Chemistry, DFT, PyTorch

- Designed pretraining on electron-density-matrix tasks so that molecular graph embeddings encode electronic structure; showed that the learned representations transfer to electron-density-related properties at both the atom and molecule level.
- On protein–ligand interaction benchmarks, outperformed ECFP and physicochemical descriptors on both binding classification and affinity prediction; currently extending the approach to electron-aware protein embeddings to model interactions more accurately.

MEVD: Multi-Epitope Vaccine Design Pipeline for LLM Agents — May 2026 – Present
FastMCP, Bayesian Optimization

- Built an end-to-end design pipeline and **exposed it as Model Context Protocol (MCP) tools** so an LLM agent can orchestrate the whole workflow (also deployed as a web service).
- Added Bayesian optimization over epitope ordering, recovering 82.4% of the attainable solubility ceiling versus 56.5% for equal-budget random search.

AI-Based Ligand Generation for Targeted Radiopharmaceuticals in Refractory Prostate Cancer — Apr 2026 – Present
Generative Design, Docking, Cosolvent MD

- Sole hands-on researcher for AI-driven candidate design in an NRF project with Asan Medical Center (No. N01261333).
- Designed dual-target binders and generated *de novo* binder candidates for a second tumor-associated antigen.

***M. tuberculosis* Inhibitor Virtual Screening** — Jul 2025 – Nov 2025
Enamine REAL, Molecular Docking

- Screened the Enamine REAL library to propose novel inhibitor candidates against a previously unexplored antibacterial target.

EXPERIENCE

Lab GPU Cluster Administrator — Feb 2026 – Present
Bioinformatics and Computational Biology Lab, KAIST

- Administer the lab's multi-node Linux GPU cluster; introduced and maintain SLURM scheduling with per-user GPU and walltime policies, and centralized Prometheus/Grafana monitoring across nodes.

Instructor, Drug Discovery with AI Track — Aug 2025 – Present
Korea Pharmaceutical and Bio-Pharma Manufacturers Association

- Taught QSAR from RDKit basics to ML and deep-learning pIC50 models, and showed how they fit into a drug discovery pipeline.
- Using agentic AI, trained medicinal chemists with little or no coding experience to run a range of models and tools on their own.

Undergraduate Research Intern — Mar 2022 – Dec 2023
Bio-organic Lab, Seoul National University

- Synthesized cell-penetrating peptides by SPPS, purified by HPLC, and quantified cellular uptake by flow cytometry.

PUBLICATIONS & PRESENTATIONS

[1] **D. H. Kim** and D. S. Kim, "DeMiR: Density-Matrix-Informed Representations for Molecular Property Prediction," **ICLR 2027 (under review)**.

[2] **D. H. Kim**, S. J. Choi, S. M. Lee, and D. S. Kim, "MEVD: An integrated multi-epitope vaccine design pipeline exposed as Model Context Protocol tools for LLM agents," *BIOINFO/GIW ISCB-Asia 2026*, **poster (accepted Sep 2026).**

[3] **D. H. Kim** and D. S. Kim, "Improving Protein–Ligand Interaction Prediction using Electron-Aware Embeddings," *BIOINFO 2025* **(Excellent Poster Award)**.

TECHNICAL SKILLS

Languages	Python (primary), Bash, R
ML / Deep Learning	PyTorch, PyG, e3nn; equivariant GNNs (eSCN, MACE, Equiformer); diffusion, flow matching, GFlowNet; Bayesian optimization
LLM Systems	Model Context Protocol (FastMCP), Anthropic API / Admin API, agent-driven scientific pipelines
Software	RDKit, GROMACS, DFT, docking (gnina, AutoDock), Protein Folding Models (AlphaFold3, Boltz-2, ...)
HPC / Infrastructure	Linux, SLURM, CUDA toolchain and multi-GPU training, Docker, NFS, Prometheus / Grafana
Languages (spoken)	Korean (native), English (fluent)